



INTERNSHIP REPORT
FIBER OPTIC COMMUNICATIONS, FTTH NETWORK
ARCHITECTURE
AND HOME APPLIANCES MAINTENANCE

Submitted by:

FATHIMA DILNA KP

Under the Faculty Guidance of:

Dr. Mohemmed Shanid N A

DEPARTMENT OF PHYSICS

SULLAMUSSALAM SCIENCE COLLEGE, AREEKODE

CERTIFICATE

This is to certify that the internship report entitled "Fiber Optic Communications, FTTH Network Architecture and Electrical Home Appliances Maintenance" is a bona fide record of the industrial training carried out by FATHIMA DILNA KP under my academic supervision and guidance, in corporate collaboration with Bfone Fibernet, Edavannappara, in partial fulfillment of the requirements for the award of the Degree of Bachelor of Science in Physics.

Dr. Mohemmed Shanid N A
Head of the Department
Department of Physics

External Examiner

Internal Examiner

CONTENTS

CERTIFICATE.....	2
INTERNSHIP ACTIVITY SCHEDULE.....	5
Part I: Fiber Optic Communications and FTTH Network Architecture.....	6
Part Introduction: Fiber Optic Infrastructure.....	7
1 Introduction to Optical Fiber Communication.....	7
1.1 Overview.....	7
1.2 The Basic Fiber Optic Link.....	7
1.3 Advantages of Fiber Optics.....	7
2 Principle and Cable Structure.....	9
2.1 Principle of Operation: Total Internal Reflection.....	9
2.2 Cable Anatomy.....	9
2.3 Cable Types.....	9
3 Link Budgeting, Splicing, and Mathematical Calculations.....	10
3.1 OFC Splicing.....	10
3.2 Logarithmic Conversion Mechanics.....	10
3.3 Splitter Loss Calculations.....	10
3.4 Standard Loss Values for Link Performance.....	10
3.5 Mathematical Verification: Power Link Budget Case Study.....	10
3.5.1 Case Study Parameter Blueprint.....	11
3.5.2 Link Status Analysis.....	11
4 Optical Fiber Splicing: Detailed Analysis.....	12
4.1 What is Splicing?.....	12
4.2 Types of Splicing.....	12
4.2.1 Fusion Splicing.....	12
4.2.2 Mechanical Splicing.....	12
4.3 Importance and Significance.....	12
4.4 Step-by-Step Fusion Splicing Process.....	12
4.5 Advantages.....	13
4.6 Future Scope.....	13
4.7 Conclusion.....	13
5 Introduction to FTTH & Passive Optical Networks (PON).....	14
5.1 Overview of FTTH.....	14
5.2 The FTTH Network Architecture Layers.....	14
5.2.1 A. The Feeder Segment.....	14
5.2.2 B. The Distribution Segment.....	14
5.2.3 C. The Drop Segment.....	14
5.3 Core Hardware Components.....	14
5.3.1 Active Components (Power Required).....	14

5.3.2 Passive Components (No Power Required).....	14
6 FTTH Deployment, Coding and Testing.....	16
6.1 Fiber Color Coding Standard (TIA-598-C).....	16
6.2 Physical Deployment & Site Survey.....	16
6.2.1 Site Survey & Planning.....	16
6.2.2 Outdoor Routing Operations.....	16
6.2.3 Indoor Termination.....	16
6.3 Splicing Methods Comparison.....	16
6.4 Link Budget & Testing Benchmarks.....	17
6.4.1 Wavelength Multiplexing.....	17
6.4.2 Testing Parameters.....	17
6.5 Troubleshooting & Indicators.....	17
7 Field Visit: BSNL Exchange Office, Manjeri.....	19
7.1 Introduction to the Field Visit.....	19
7.2 Observations and Core Infrastructure.....	19
7.3 Conclusion of the Visit.....	19
Part Conclusion: Fiber Optic Infrastructure.....	19
Part II: Electrical Home Appliances Maintenance and Troubleshooting.....	20
Part Introduction: Domestic Electrical Engineering.....	21
8 Fundamentals of Electrical Systems & Safety.....	21
8.1 Introduction.....	21
8.2 Fundamentals of Electrical Systems & Safety Overview.....	21
8.2.1 Circuit Behaviors & Laws.....	21
8.2.2 Earthing and Protection Systems.....	22
9 Technical Breakdown & Diagnostics of Core Home Appliances.....	23
9.1 Electric Iron.....	23
9.2 Induction Cooker.....	23
9.3 Mixer Grinder.....	24
9.3.1 Ceiling Fan.....	24
9.3.2 Wall Fan.....	24
Chapter 10: Troubleshooting & Maintenance Practice Matrix.....	26
10.1 Internship Outcomes & Skills Acquired.....	27
11 Applied Practical Calculations.....	28
11.1 Domestic Electricity Consumption Auditing.....	28
11.2 Charge Calculation in Audio Systems.....	28
Part Conclusion: Domestic Electrical Engineering.....	28

INTERNSHIP ACTIVITY SCHEDULE

Day	Date	Topics / Activities Covered
Day 1	21.05.2026	Optical Fiber Theory Class (Introduction to OFC)
Day 2	22.05.2026	Splicing Practical Section
Day 3	23.05.2026	Repairing of the Ceiling Fan
Day 4	25.05.2026	• Theory Class of Fiber To The Home (FTTH) • Repairing of Mixer Grinder • Field Visit of Splicing
Day 5	26.05.2026	Field Visit at Manjeri Telephone Exchange
Day 6	08.06.2026	Power Calculation Theory Class
Day 7	09.06.2026	• Repairing of Iron Box • Repairing of Wall Fan • Repairing of Induction Cooker

Table 1: Day-wise Internship Activity Log

Part I

Fiber Optic Communications and FTTH Network Architecture

PART INTRODUCTION: FIBER OPTIC INFRASTRUCTURE

Today's telecommunication sector is heavily reliant on high-speed data transmission networks built on principles of optical physics. Moving beyond legacy copper-wire networks, optical fibers guide light signals to transport large amounts of data over vast distances with minimal signal degradation. This section of the report details a technical analysis of modern Fiber-Optic Communication (OFC) and Fiber-to-the-Home (FTTH) setups, based on practical experience at Bfone Fibernet. The core purpose here is to outline the entire journey of the light signal, from the hardware source down to the home ONT. The chapters in this section study Total Internal Reflection (TIR) guiding light in glass cores, multi-core cable design, and the calculations needed for fiber link budgets. Moreover, we document standard field procedures, including precision splicing, color code identification, and troubleshooting techniques used by technicians. This section finishes with a technical summary of our field study at the Manjeri BSNL Exchange Office.

CHAPTER 1

INTRODUCTION TO OPTICAL FIBER COMMUNICATION

1.1 Overview

Fiber-optic communication (OFC) involves transmitting data from a source to a destination by sending light pulses through glass or plastic fibers. In this system, light acts as an electromagnetic carrier wave modulated to carry digital and analog data.

1.2 The Basic Fiber Optic Link

The basic structure of a fiber-optic communication link includes several essential elements:

- Encoder/Source: Takes input data and translates it into digital electrical pulses.
- Transmitter (E/O): Uses a semiconductor LED or laser diode to convert electrical pulses into light signals.
- Optical Fiber: The silica glass or plastic medium that guides the light pulse over a distance.
- Receiver (O/E): Employs a photodiode (PIN or APD) to detect the light signal and convert it back to electrical form.
- Decoder: Reconstructs the original digital data from the electrical pulses.

1.3 Advantages of Fiber Optics

- Non-conductive: Made of glass/plastic, eliminating electrical spark risks and grounding problems.
- Electromagnetic Immunity: Unaffected by electromagnetic interference (EMI), radio frequency interference, or lightning strikes.
- Large Bandwidth: Capable of supporting incredibly high data rates and channel capacities.
- Low Loss: Extremely low signal attenuation, allowing long spans without repeaters.

- **Small and Lightweight:** Thin profile and lightweight nature simplify installation in conduits and aerial runs.

CHAPTER 2

PRINCIPLE AND CABLE STRUCTURE

2.1 Principle of Operation: Total Internal Reflection

The propagation of light along the fiber core is governed by the physical principle of Total Internal Reflection (TIR). The occurrence of TIR requires two strict conditions:

- The core material must have a higher refractive index (n_1) than the surrounding cladding (n_2).
- The light must strike the core-cladding boundary at an angle of incidence exceeding the critical angle (θ_c).

The mathematical formula for the critical angle is:

$$\theta_c = \sin^{-1}(n_2 / n_1) \quad (2.1)$$

2.2 Cable Anatomy

A typical optical fiber cable is constructed with the following layers from the inside out:

- Core: The central glass or plastic channel through which the light pulses travel.
- Cladding: The surrounding layer of glass with a lower refractive index that keeps the light trapped in the core.
- Buffer Coating: A plastic layer surrounding the cladding to shield it from moisture and mechanical stress.
- Strength Member: High-tensile aramid yarn (Kevlar) or steel wire to absorb tension during pulling.
- Outer Jacket: The tough plastic outer sheath (PVC or PE) protecting the cable from environmental factors.

2.3 Cable Types

- Tight-Buffer Cable: Features thick buffer coating directly on the fiber, ideal for indoor routing and patch cords.
- Loose-Tube Cable: Fibers lie loose inside gel-filled tubes, providing excellent protection against water and temperature shifts in outdoor ducts.
- Ribbon Cable: Multiple fibers aligned in flat rows, maximizing fiber density in tight spaces.

CHAPTER 3

LINK BUDGETING, SPLICING, AND MATHEMATICAL CALCULATIONS

3.1 OFC Splicing

Fiber splicing refers to creating permanent joins between two fiber cores. The primary techniques include:

- Fusion Splicing: Fusing the two glass ends together using a high-voltage electric arc.
- Mechanical Splicing: Aligning fiber cores within a precision alignment fixture and securing them.

3.2 Logarithmic Conversion Mechanics

The logarithmic decibel-milliwatt (dBm) scale is standard in fiber optics to simplify power loss calculations across links:

$$P_{\text{dBm}} = 10 \log_{10}(P_{\text{mW}} / 1 \text{ mW}) \quad (3.1)$$

3.3 Splitter Loss Calculations

The ideal insertion loss of a passive optical splitter depends directly on the split ratio or output ports (N):

$$\text{Loss}_{\text{Ideal}} = 10 \log_{10}(N) \quad (3.2)$$

3.4 Standard Loss Values for Link Performance

The following standard loss values are utilized by technicians to calculate link budgets:

Network Element Component	Standard Loss Value
Fiber Path Loss @ 1310 nm / 1490 nm	0.40 dB per km
Fiber Path Loss @ 1550 nm	0.25 dB per km
Fusion Splice Junction	0.10 dB per splice
SC/UPC Physical Connector Pair	0.35 dB per connector
SC/APC Physical Connector Pair	0.25 dB per connector
1 × 2 Splitter Module	3.0 dB average insertion loss
1 × 16 Splitter Module	13.0 dB average insertion loss
1 × 32 Splitter Module	16.0 dB average insertion loss

Table 3.1: Standard Loss Parameters Used in FTTH Networks

3.5 Mathematical Verification: Power Link Budget Case Study

$$P_r = P_t - \text{Loss}_{\text{total}} \quad (3.3)$$

3.5.1 Case Study Parameter Blueprint

Let us examine a 7 km optical fiber link operated by Bfone Fibernet. This specific network span contains 4 fusion splices, 2 SC/UPC connectors, and two splitters in series: a 1×16 and a 1×8 splitter. The loss budget details for this specific fiber span are calculated below:

- Fiber line loss: 7 km of fiber at 0.40 dB/km results in a loss of 2.8 dB
- First splitter stage (1×16) loss: 13.0 dB
- Second splitter stage (1×8) loss: 10.0 dB
- Fusion splice losses: 4 splices at 0.10 dB each equals 0.4 dB
- Connector losses: 2 SC/UPC pairs at 0.35 dB each equals 0.7 dB

The total insertion loss is calculated by adding the individual attenuation values:

$$\begin{aligned}\text{Total Attenuation Loss} &= 2.8 + 13.0 + 10.0 + 0.4 + 0.7 \\ &= 26.9 \text{ dB}\end{aligned}$$

3.5.2 Link Status Analysis

For reliable performance, typical ONT optical receivers must receive light power in the range of -10 dBm to -28 dBm: Minimum Receiver Threshold ≤ -28 dBm

$$\text{Receiver Sensitivity Threshold} \leq -28 \text{ dBm} \quad (3.4)$$

Since the total attenuation is 26.9 dB, assuming an OLT transmitter power of 0 dBm, the received signal power is -26.9 dBm. This is above the receiver threshold of -28 dBm, confirming that the fiber link is viable and will operate reliably.

CHAPTER 4

OPTICAL FIBER SPLICING: DETAILED ANALYSIS

4.1 What is Splicing?

Fiber splicing is defined as the technique of permanently connecting two optical fiber ends to form a seamless light path. While connectors are built for quick disconnects, splices are permanent joints optimized for stability. The main objective of splicing is to minimize insertion loss and keep return loss (back-reflection) low.

4.2 Types of Splicing

The two principal splicing techniques are:

4.2.1 Fusion Splicing

Fusion Splicing is the standard technique in telecom networks. It uses a controlled electric arc to melt and fuse the glass fiber ends. Attenuation: Usually between 0.01 dB and 0.05 dB. Applications: Used in long-distance backbones and residential FTTH deployments.

4.2.2 Mechanical Splicing

Mechanical splicing aligns the two fiber ends inside a precision sleeve with index-matching gel to bridge the air gap. Attenuation: Usually ranges from 0.2 dB to 0.5 dB. Applications: Temporary fiber joins, emergency field repairs, and low-cost terminations.

4.3 Importance and Significance

- Link Extension: Joins shorter fiber spools to build long-distance network runs.
- Signal Integrity: Assures optimal light propagation and very low insertion loss.
- Back-reflection Control: Limits return loss (reflections) to protect high-power laser transmitters.
- Durability: Creates robust, long-lasting joints that resist temperature and physical strain.

4.4 Step-by-Step Fusion Splicing Process

1. Stripping: Stripping away the 250 μm polymer buffer coating using precision fiber strippers to expose bare glass.
2. Cleaning: Cleaning the bare glass surface with 99% pure Isopropyl Alcohol (IPA) and lint-free wipes.
3. Cleaving: Cutting the fiber using a precision cleaver to achieve a flat 90-degree angle (less than 0.5 degrees deviation).
4. Fusion: Aligning the fibers automatically using PAS and applying a localized electric arc to fuse the glass at 2000°C.
5. Protection: Sliding a heat-shrink protector sleeve over the splice point and baking it in the splicer's oven to seal it.

4.5 Advantages

- Provides the absolute lowest signal attenuation (< 0.05 dB).
- Forms a highly stable, permanent glass-to-glass bond.
- Unbothered by mechanical vibrations and thermal expansion.
- Very compact, fitting easily into splice trays and closures.

4.6 Future Scope

- Automation: Introduction of automated splicing machines and robotics for harsh environments.
- Multi-core Fiber Splicing: Upgrading equipment to splice multi-core fibers that support higher spatial bandwidth.
- Enhanced Precision: Incorporating nanometer-level alignment to reach near-zero signal loss in critical paths.

4.7 Conclusion

In summary, splicing is fundamental to building reliable optical infrastructure. Creating permanent, low-loss connections ensures that gigabit data speeds are maintained over long spans. Fusion splicing remains the gold standard, while mechanical connectors provide critical utility for quick repairs.

CHAPTER 5

INTRODUCTION TO FTTH & PASSIVE OPTICAL NETWORKS (PON)

5.1 Overview of FTTH

Fiber-to-the-Home (FTTH) is the modern standard for high-speed internet delivery, replacing legacy copper telephone and cable lines completely with fiber lines run directly to consumer homes. The dominant technology in FTTH is PON, or Passive Optical Network. It is 'passive' because there are no powered switches in the field; light is divided using glass splitters. This design reduces power consumption, cuts down on hardware failures, and lowers operating costs. Through this point-to-multipoint topology, a single fiber feeds multiple households.

5.2 The FTTH Network Architecture Layers

The physical deployment of FTTH networks consists of three main segments:

- A. The Feeder Segment: Connects the Central Office (CO) to the Fiber Distribution Hub (FDH). It uses heavy, high-fiber-count cables (48 to 144+ cores) to carry signals over long trunk routes (up to 20 km) without splitting.
- B. The Distribution Segment: Spans from the FDH to street-level boxes like the Optical Distribution Point (ODP). Feeder fibers are split at the FDH, and distribution cables (12 to 24 cores) run along streets on poles or in ducts.
- C. The Drop Segment: The final stretch running from the ODP to the customer's ONT. It uses a single or dual-core drop cable designed with high bend tolerance (G.657 standard) to navigate tight corners inside home conduits.

5.3 Core Hardware Components

The architecture is built on a close cooperation between powered active components and non-powered passive components:

5.3.1 Active Components (Power Required)

OLT (Optical Line Terminal): The central switch situated at the Central Office. It translates backbone internet traffic into optical pulses and coordinates downstream/upstream transmission scheduling.

ONT / ONU (Optical Network Terminal / Unit): The subscriber-end modem. It converts the incoming light signals from the drop cable into standard electrical Ethernet or Wi-Fi signals for household use.

5.3.2 Passive Components (No Power Required)

Optical Splitters: Passive glass components that divide a single input light wave into multiple output paths (e.g., 1:8, 1:16, 1:32) to feed multiple subscribers without power.

FDH / ODF Cabinets: Weatherproof enclosures hosting splitters and splice trays, securing connections against dust, water, and heat.

CHAPTER 6

FTTH DEPLOYMENT, CODING AND TESTING

6.1 Fiber Color Coding Standard (TIA-598-C)

To identify individual fibers in high-core-count cables, the industry uses a standardized 12-color coding sequence (TIA-598-C). For larger cables, the color sequence repeats using a black stripe or tracer.

No.	Fiber Color	Symbol	No.	Fiber Color	Symbol
1	Blue	BL	7	Red	RD
2	Orange	OR	8	Black	BK
3	Green	GR	9	Yellow	YL
4	Brown	BR	10	Violet	VI
5	Slate (Grey)	SL	11	Rose (Pink)	RS
6	White	WH	12	Aqua	AQ

Table 6.1: TIA-598-C Color Coding Identification

6.2 Physical Deployment & Site Survey

6.2.1 Site Survey & Planning

Prior to cable deployment, technicians conduct a site survey covering these steps:

- Route Mapping: Planning the shortest cable path from the street ODP to the customer premises.
- Structural Assessment: Finding entry points and deciding on the best indoor ONT location.
- Safety Check: Checking for power lines to maintain safe spacing and avoid electrical hazards.
- BOM Calculation: Creating the Bill of Materials including clamps, drop cables, and wall outlets.

6.2.2 Outdoor Routing Operations

Aerial Deployment: Drop wires are suspended between poles using drop clamps. Technicians leave slack loops to accommodate expansion and contraction from temperature changes. Underground Deployment: In suburban areas, fiber is pulled through buried protective HDPE ducts to shield it from excavation and weather damage.

6.2.3 Indoor Termination

The fiber cable enters the home and is terminated at a wall-mounted fiber outlet or rosette box. A bend-insensitive patch cord (G.657 standard) is then run from the outlet to the ONT, allowing tight bends around walls without causing signal loss.

6.3 Splicing Methods Comparison

Feature	Fusion Splicing	Mechanical / Fast Connectors
---------	-----------------	------------------------------

Mechanism	Uses a high-voltage electric arc to melt and fuse glass tips permanently.	Physically aligns two fiber ends inside a sleeve using index-matching gel.
Signal Loss	Extremely low (< 0.05 dB to 0.1 dB).	Slightly higher (0.2 dB to 0.5 dB).
Pros & Cons	Most reliable, but requires an expensive fusion splicer machine.	Fast setup, low initial tool cost. Ideal for home drop termination.

Table 6.2: Comparison of Fusion and Mechanical Splicing

6.4 Link Budget & Testing Benchmarks

Single-mode fiber supports bidirectional traffic by using separate wavelengths for upstream and downstream data:

6.4.1 Wavelength Multiplexing

- 1490 nm: Carries downstream internet and voice traffic from OLT to subscriber ONT.
- 1310 nm: Used for upstream traffic uploads from ONT to the central OLT.
- 1550 nm: Allocated for downstream analog or digital TV overlays.

6.4.2 Testing Parameters

- Optical Power Meter (OPM): Used to measure the optical power level at the ONT. The normal range is -15 dBm to -25 dBm. Levels below -27 dBm indicate an attenuation fault.
- OTDR (Optical Time Domain Reflectometer): Sends light pulses down the fiber to analyze backscatter, mapping splice points and finding the exact location of cable cuts.

6.5 Troubleshooting & Indicators

Technicians diagnose residential link faults by reading the status LEDs on the ONT:

ONT Indicator	Status State	Meaning / Corrective Action
LOS (Red)	Steady Glow	Loss of Signal: Total physical fiber cut or severe macro-bend. Check outside plant with OTDR.
PON (Green)	Blinking / Flashing	Registering: Authentication in progress. If persistent, light power is weak or configuration profile is missing.
PON (Green)	Steady Solid Glow	Connected: Symmetrical operational link established successfully.
Power Levels	Below -27 dBm	Power Fault: Dirty connector faces or high-loss splice. Rework connection or clean with a OneClick cleaner.

Table 6.3: ONT Troubleshooting Matrix

CHAPTER 7

FIELD VISIT: BSNL EXCHANGE OFFICE, MANJERI

7.1 Introduction to the Field Visit

To gain practical exposure to telecom systems, we visited the BSNL Telephone Exchange in Manjeri. This field visit provided a real-world look at central office operations. This excursion let us trace the physical flow of data from central carrier routers to the local distribution loops.

7.2 Observations and Core Infrastructure

Technicians at the exchange demonstrated the OLT racks, optical distribution frames (ODF), and multi-fiber termination racks. They explained how incoming voice and data feeds are combined and routed onto fiber cores. The OLT works in tandem with key backend routing and authentication nodes: The Broadband Network Gateway (BNG) manages subscriber sessions, allocates dynamic/static IP addresses, and controls user bandwidth. The MNGN platform provides carrier-grade packet transport, integrating FTTH broadband, IPTV, and business data links onto a unified network. The OLT connects to customer ONTs via the optical distribution network.

7.3 Conclusion of the Visit

This technical tour was highly educational, helping us visualize how OLTs and BNGs operate in a real carrier network. It successfully linked our classroom physics to modern telecom infrastructure.

PART CONCLUSION: FIBER OPTIC INFRASTRUCTURE

To summarize, this first section has covered the core architectural principles of modern fiber-optic communications. We studied light propagation in single-mode cores and analyzed fiber structure and attenuation in distribution cables. Our study of optical splitters and link budget calculations showed how networks maintain signal strength for thousands of users. Practical exercises at Bfone Fibernet allowed us to master fusion splicing, color code identification, and OTDR troubleshooting. Additionally, the BSNL Exchange visit illustrated macro-scale network architecture, showing the interaction between OLTs, BNG routers, and ONTs. Ultimately, this section showed that reliable gigabit broadband requires both accurate math and high-quality field installation.

Part II

Electrical Home Appliances

Maintenance and Troubleshooting

PART INTRODUCTION: DOMESTIC ELECTRICAL ENGINEERING

Electro-mechanical appliances are essential to the comfort and function of modern households. Devices utilizing resistive heating, induction cooking, or single-phase induction motors apply core principles of thermodynamics and electromagnetism. This section of the report documents the technical details, troubleshooting steps, and maintenance practices studied during our training. The goal was to build a solid diagnostic mindset based on electrical theory rather than basic parts-swapping. The following chapters cover domestic wiring, series and parallel circuits, and safety devices like grounding, MCBs, and RCCBs that protect against shocks and short circuits. We then detail the construction and troubleshooting of seven appliances: the electric iron, induction cooker, mixer grinder, water geyser, ceiling fan, wall fan, and utility pump. Finally, this module presents practical math problems on electricity auditing and electrical charge calculations.

CHAPTER 8 FUNDAMENTALS OF ELECTRICAL SYSTEMS & SAFETY

8.1 Introduction

Household appliances are central to modern life, running constantly to support cooking, ventilation, and water supply. Over time, heavy use causes mechanical wear, insulation breakdowns, and electrical faults. This module was completed to apply electrical theory to practical repair. The course covered safety protocols, wiring layouts, and the step-by-step repair of seven core home appliances. Our goal was to analyze the physics of these appliances—like resistive heating, electromagnetism, and motor action—while practicing safe work habits.

8.2 Fundamentals of Electrical Systems & Safety Overview

Prior to starting repairs, we studied basic circuit behaviors and safety components:

8.2.1 Circuit Behaviors & Laws

- Ohm's Law & Power: We use $V = IR$ and $P = VI$ to calculate normal resistance values, allowing us to find short circuits and open windings during testing.
- Series Wiring: Current is constant while voltage drops. Used for safety switches, fuses, and thermostats to interrupt current flow.
- Parallel Wiring: Voltage remains constant while current divides. Standard for home wiring to ensure appliances operate independently.

$$V = IR \quad \text{and} \quad P = VI = I^2R \quad (8.1)$$

8.2.2 Earthing and Protection Systems

- Earthing: Connects the metal body of an appliance to the ground. If a hot wire touches the chassis, current flows to earth, preventing shock to the user.
- Fuses & MCBs: Devices that open the circuit during short circuits or overloads, protecting wiring from overheating.
- RCCB / ELCB: Detects small leakages (30mA) to ground by comparing live and neutral currents, tripping instantly to prevent severe shock.

CHAPTER 9

TECHNICAL BREAKDOWN & DIAGNOSTICS OF CORE HOME APPLIANCES

9.1 Electric Iron

A. Working Principle

The electric iron operates on Joule's Heating Effect. Current flowing through a high-resistance heating element generates heat proportional to the resistance and square of the current ($H = I^2Rt$). A bimetallic strip thermostat controls the heat. Since the two bonded metals expand at different rates, heating bends the strip to open contacts and cut the circuit. Cooling resets the strip, restarting the heating cycle.

$$H = I^2Rt \quad (9.1)$$

B. Key Components and Parts

- Soleplate: A smooth, heavy metal plate (aluminum/cast iron) that transfers heat to the fabric.
- Heating Element: A nichrome wire ribbon wrapped in mica sheets to insulate the live voltage while conducting heat.
- Bimetallic Thermostat: The temperature regulator with an adjustable dial that sets the contact gap.
- Thermal Fuse: A safety fuse that melts permanently if temperatures exceed safe limits due to thermostat failure.

C. Appliance Conclusion

Studying the iron showed the practical side of Joule heating and thermal feedback loops. Working on it highlighted the role of thermal fuses in protecting against fire hazards.

9.2 Induction Cooker

A. Working Principle

Unlike standard stoves that use heat conduction, induction cooktops operate using electromagnetic induction and eddy currents. An internal inverter converts 50Hz AC mains into high-frequency AC (20-40kHz). This current flows through an induction coil to generate an alternating magnetic field. Placing a ferromagnetic pan on the glass induces eddy currents in the pan's base. The pan's electrical resistance causes it to heat up directly, leaving the glass top relatively cool.

B. Key Components and Parts

- Induction Coil: A flat, spiral copper coil that creates the alternating magnetic field.
- IGBT: A high-power semiconductor switch that drives the high-frequency inverter circuit.
- Main PCB & Control Unit: The controller that processes settings, manages power level switching, and monitors safety sensors.
- Fan & Temperature Sensor (RTD): A cooling fan and thermistor that prevent overheating and detect boil-dry conditions.

C. Appliance Conclusion

Induction stove troubleshooting offered great experience with solid-state inverters and magnetic heating. It showed the importance of tracing faults at the board component level.

9.3 Mixer Grinder

A. Working Principle

Mixer grinders use an AC Universal Motor to convert electrical energy into rotation. The universal motor is series-wound and runs on either AC or DC. Current flows through the stator field coils and the rotor windings through carbon brushes. The resulting magnetic fields produce high starting torque and speeds up to 20,000 RPM, ideal for grinding and blending.

B. Key Components and Parts

- Stator & Armature: The stationary field coils and rotating core that convert electrical energy to torque.
- Commutator & Brushes: Sliding carbon contacts that deliver current to the rotating commutator bars.

9.3.1 Ceiling Fan

A. Working Principle

Ceiling fans utilize a Single-Phase Permanent Split Capacitor Induction Motor. Since single-phase AC cannot self-start a motor, the stator features main (running) and auxiliary (starting) windings. A start/run capacitor in series with the starting winding shifts the current phase by 90° . This phase difference creates a rotating magnetic field, inducing current in the rotor and turning the fan blades.

B. Key Components and Parts

- Stator & Rotor: An inner stationary core with copper windings and an outer rotating steel shell.
- Run Capacitor: A 2.25 to 2.5 microfarad capacitor that creates the phase shift to start the motor.
- Ball Bearings: Dual bearings that support the rotor shell, minimizing noise and friction.
- Down-Rod: The steel pipe assembly that suspends the fan safely from the ceiling hanger.

C. Appliance Conclusion

Repairing ceiling fans helped us understand phase-splitting in single-phase motors. The troubleshooting process showed how to distinguish winding issues from mechanical bearing wear.

9.3.2 Wall Fan

A. Working Principle

Wall fans also use a single-phase capacitor induction motor but add mechanics for oscillation. The main motor drives the blades, while a mechanical gear linkage or a small secondary motor rotates the fan assembly side-to-side. Fan speed is controlled by switching between motor winding taps or using a TRIAC circuit to vary voltage.

B. Key Components and Parts

- Blades & Grille: Curved plastic/metal blades protected by a wire mesh safety guard.

- Oscillation Gearbox: A reduction gear assembly that converts high-speed motor rotation into a slow swiveling motion.
- Speed Selector: A mechanical rotary switch, pull-cord, or electronic circuit that adjusts voltage to regulate speed.
- Thermal Cut-Off (TCO): A thermal fuse inside the stator coils that cuts power if the motor overheats due to a jam.

C. Appliance Conclusion

Working on the wall fan showed how mechanical gearboxes connect to electrical motors. It was a good lesson in motor winding safety fuses and fixing mechanical oscillation failures.

CHAPTER 10

TROUBLESHOOTING & MAINTENANCE PRACTICE MATRIX

ONT Indicator	Status State	Meaning / Corrective Action
LOS (Red)	Steady Glow	Loss of Signal: Total physical fiber cut or severe macro-bend. Check outside plant with OTDR.
PON (Green)	Blinking / Flashing	Registering: Authentication in progress. If persistent, light power is weak or configuration profile is missing.
PON (Green)	Steady Solid Glow	Connected: Symmetrical operational link established successfully.
Power Levels	Below -27 dBm	Power Fault: Dirty connector faces or high-loss splice. Rework connection or clean with a OneClick cleaner.

Table 6.3: ONT Troubleshooting Matrix

Appliance	Common Symptom	Root Cause Detected	Corrective Action Performed
Electric Iron	No heating; indicator light off	Open circuit in thermal fuse / Broken power cord	Replaced thermal fuse; trimmed and re-terminated power cord.
Induction Cooker	Dead display; won't power up	Blown main fuse due to shorted IGBT	Replaced the shorted IGBT and main glass fuse; checked gate driver circuit.
Mixer Grinder	Sparks from bottom; sudden stop	Worn carbon brushes / Overload switch tripped	Replaced carbon brushes; cleared jar jam and reset the bottom OLP switch.
Water Heater	Water not heating; indicator on	Scale buildup or open-circuit in heating element	Drained tank, removed limescale, and replaced the immersion heating element.
Ceiling Fan	Humming sound; spins very slowly	Degraded or leaked capacitor	Desoldered/disconnected old capacitor, verified value, and installed new 2.5 μ F unit.
Wall Fan	Fan rotates but does not oscillate / Overheating	Stripped linkage gears / Dry rotor shaft	Replaced the plastic gear wheel assembly; cleaned and lubricated the rotor shaft with machine oil.
Pump Motor	Motor hums but does not rotate	Seized bearings / Defective starter capacitor	Freed impeller manually; replaced bearings and

capacitor.

Table 10.1: Appliance Troubleshooting & Maintenance Practice Matrix

10.1 Internship Outcomes & Skills Acquired

- **Safety First Mentality:** Learned to always check for live voltage using a test pen, work on rubber mats, and verify ground lines before powering up devices.
- **Structured Troubleshooting:** Mastered the process of tracing faults along the circuit path, starting from the plug, checking fuses, and then testing switches and windings.
- **Multimeter Proficiency:** Gained hands-on experience using multimeters to test continuity, winding resistances, and checking live AC voltages safely.

CHAPTER 11

APPLIED PRACTICAL CALCULATIONS

11.1 Domestic Electricity Consumption Auditing

Domestic electricity usage is measured in kilowatt-hours (kWh), with one unit of electricity equal to 1 kWh.

Calculate the 30-day energy consumption for a home operating two 40W tubes for 6 hours/day, three 65W fans for 8 hours/day, and auxiliary loads of 150W for 2 hours/day.

$$E_{\text{Lights}} = 2 \times 40 \text{ W} \times 6 \text{ hrs/day} \times 30 \text{ days} = 14,400 \text{ Wh} = 14.4 \text{ kWh}$$

$$E_{\text{Fans}} = 3 \times 65 \text{ W} \times 8 \text{ hrs/day} \times 30 \text{ days} = 46,800 \text{ Wh} = 46.8 \text{ kWh}$$

$$E_{\text{Misc}} = 150 \text{ W} \times 2 \text{ hrs/day} \times 30 \text{ days} = 9,000 \text{ Wh} = 9.0 \text{ kWh}$$

$$E_{\text{Total}} = 14.4 \text{ kWh} + 46.8 \text{ kWh} + 9.0 \text{ kWh} = 70.2 \text{ kWh (Units)}$$

11.2 Charge Calculation in Audio Systems

An appliance connected to a 230V line consumes 8 Megajoules (MJ) of energy. Calculate the total electric charge that passed through the system.

$$\text{Given: } V = 230 \text{ V, } E = 8 \times 10^6 \text{ J}$$

$$\text{Equation: } E = qV$$

$$q = E / V$$

$$= 8 \times 10^6 \text{ J} / 230 \text{ V}$$

$$= 3.48 \times 10^4 \text{ Coulombs (C)}$$

PART CONCLUSION: DOMESTIC ELECTRICAL ENGINEERING

In summary, this section connected classroom physics with real-world electrical troubleshooting. Studying the components and wiring of seven core appliances provided valuable insight into domestic electrical layouts. We learned to trace faults, such as open bimetallic contacts in irons or degraded start capacitors in fans, using multimeters. The exercises emphasized the importance of safety devices like thermal cut-offs (TCOs) and overload protectors (OLPs) in preventing damage. Furthermore, calculating energy consumption and charge transfer helped us understand power management. Ultimately, the hands-on repair skills and diagnostic logic learned during this course form a valuable toolkit for electrical maintenance.